# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 13: **Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The distance from the Earth to the Sun is 1.5 x 10^11 m and the speed of light is 3 x 10^8 m/s. The time taken by sunlight is: 1

**(a)** 500 s **(b)** 5000 s **(c)** 50 s **(d)** 5 s

**2.** A 1 cm rise of the sea level due to melting ice would mostly affect: 1

**(a)** deserts **(b)** polar caps only
**(c)** coastal low-lying areas **(d)** mountain tops

**3.** In the water cycle, the Sun provides energy for: 1

**(a)** run-off **(b)** condensation **(c)** evaporation **(d)** precipitation

**4.** Which of these processes removes nitrogen gas from the air for plants? 1

**(a)** respiration **(b)** transpiration
**(c)** nitrogen fixation **(d)** burning

**5.** Which of the following reduces your carbon footprint the most? 1

**(a)** using public transport and saving energy **(b)** using plastic
**(c)** burning waste **(d)** using air conditioners all day

**6.** If the Earth received 1000 W per square metre and reflected 30 per cent, the radiation absorbed is: 1

**(a)** 1030 W per square metre **(b)** 300 W per square metre
**(c)** 700 W per square metre **(d)** 970 W per square metre

**7.** The forest absorbs CO2 at 20 tonnes per hectare per year. For 50 hectares per year it absorbs: 1

**(a)** 400 tonnes **(b)** 70 tonnes **(c)** 1000 tonnes **(d)** 2.5 tonnes

**8.** The movement of water from land to sea through rivers is: 1

**(a)** evaporation **(b)** transpiration **(c)** condensation **(d)** run-off

**9.** Lightning contributes to the nitrogen cycle by: 1

**(a)** making CO2 **(b)** making ozone
**(c)** converting N2 into nitrogen oxides **(d)** forming proteins

**10.** A 5-star rated appliance is preferred because it: 1

**(a)** costs less always **(b)** uses less electricity
**(c)** is heavier **(d)** looks better

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Calculate the time taken for sunlight to reach Earth: distance 1.5 x 10^11 m and speed 3 x 10^8 m/s, in minutes. 2

**12.** Why do pulses (legumes) enrich the soil with nitrogen? 2

**13.** Why are the polar regions more affected by climate change? 2

**14.** A surface has an albedo of 0.3. Of 1000 W/m squared incident, find the reflected and absorbed power. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Explain the effect of albedo on the climate with the three surfaces: snow, forest, ocean, using values 0.85, 0.1, 0.06. 3

**16.** Explain how the human activities have disturbed the nitrogen cycle. 3

**17.** Describe how plastics affect the Earth system. 3

OR

Explain how the ozone hole forms over Antarctica in brief. 3

**18.** Describe the ways in which the oceans act as a climate regulator and a carbon sink. 3

OR

Explain how the oxygen cycle is linked with the carbon cycle. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** The Sun's radiation of 1000 W/m squared falls on a surface of 2 square metres. Find the energy received in 1 hour and the energy absorbed by
(a) snow with albedo 0.85 and
(b) forest with albedo 0.1 in kWh. 5

OR

Calculate the amount of CO2 absorbed by a forest of 100 hectares if each hectare absorbs 6 tonnes per year; compare it with the emission of 600 cars each emitting 2.4 tonnes per year. 5

**20.** Explain the nitrogen cycle with the roles of lightning, nitrogen-fixing bacteria, nitrifying bacteria, plants, animals, decomposers and denitrifying bacteria. 5

OR

Describe the water cycle's reservoirs and fluxes with the percentage values and the impact of human activity. 5

## END OF QUESTION PAPER — Total: 40 Marks

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